

Implementing HNSW with Vector Databases

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Title Slide

► Implementing HNSW with Vector Databases



Learning Objectives

- ▶ **Compare Vector Database types**
- ▶ **Install FAISS library**
- ▶ **Configure HNSW parameters**
- ▶ **Perform vector searches**



System Requirements

- ▶ **OS: Windows, Linux, or Mac**
- ▶ **Software: Python 3.8 or higher**
- ▶ **Editor: Jupyter Notebook or VS Code**

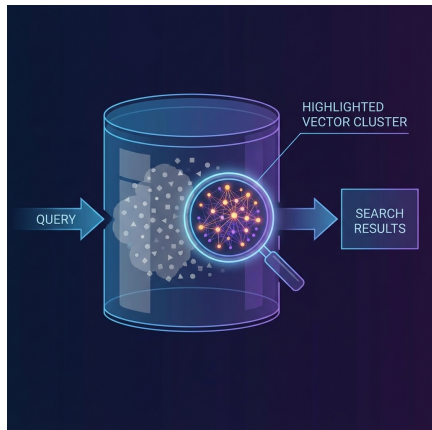


Internet



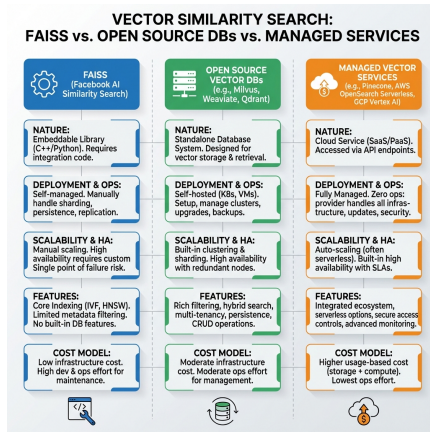
What is a Vector Database?

- ▶ **Optimized 'address book' for vectors**
- ▶ **Converts query to vector**
- ▶ **Uses HNSW to find closest neighbors**



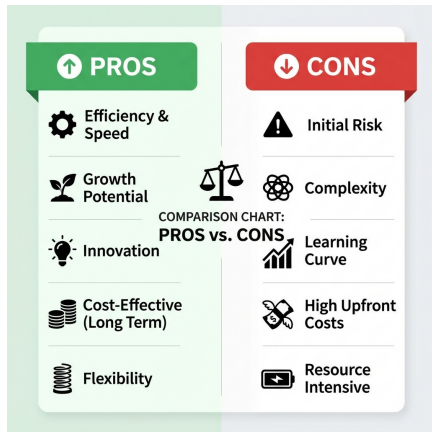
Choosing Your Library Tool

- ▶ 1. In-Memory Libraries (FAISS)
- ▶ 2. Open-Source Databases (Chroma, Weaviate)
- ▶ 3. Managed Services



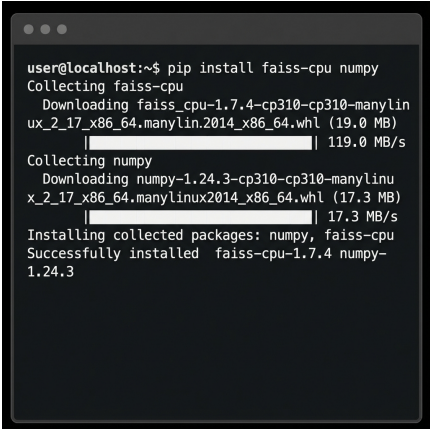
Tool Comparison

- ▶ **FAISS: Fast, local, no persistence**
- ▶ **Open-Source: Powerful, scalable, self-managed**
- ▶ **Managed: Easy, scalable, paid**



Installing FAISS

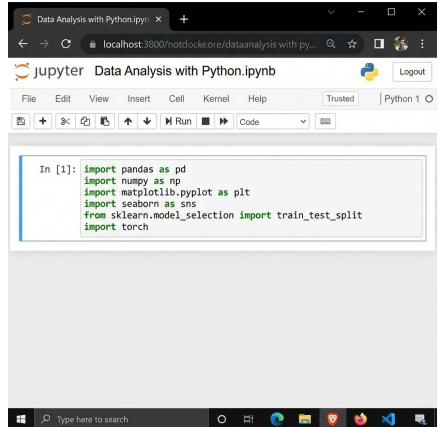
- ▶ **Step 1: Open terminal or command prompt**
- ▶ **Step 2: Run installation command**



```
user@localhost:~$ pip install faiss-cpu numpy
Collecting faiss-cpu
  Downloading faiss_cpu-1.7.4-cp310-cp310-manylin
ux_2_17_x86_64.manylinux2014_x86_64.whl (19.0 MB)
    |██████████████████████████████████████| 119.0 MB/s
Collecting numpy
  Downloading numpy-1.24.3-cp310-cp310-manylinu
x_2_17_x86_64.manylinux2014_x86_64.whl (17.3 MB)
    |██████████████████████████████████████| 17.3 MB/s
Installing collected packages: numpy, faiss-cpu
Successfully installed faiss-cpu-1.7.4 numpy-
1.24.3
```


Importing Libraries

- ▶ **Step 1: Open Python editor**
- ▶ **Step 2: Import faiss and numpy**



Setting Dimensions

- ▶ **Define vector dimension (d)**
- ▶ **Create dummy data**

Creating HNSW Index

- ▶ **Configure M (neighbors)**
- ▶ **Initialize IndexHNSWFlat**

Adding Data

- ▶ **Add vectors to index**
- ▶ **Verify count**

Configuring Search Parameters

- ▶ **Set efSearch**
- ▶ **Trade-off: Speed vs Accuracy**

Performing a Search

- ▶ **Create query vector**
- ▶ **Search for k nearest neighbors**

Summary

- ▶ **FAISS for fast prototyping**
- ▶ **Key parameters: d , M , efSearch**
- ▶ **Trade-offs in vector search**

Assignment

- ▶ **Experiment with M values**
- ▶ **Compare search results**

Thank You

- ▶ **EduPyramids Educational Services Private Limited**
- ▶ **SINE, IIT Bombay**

